
A Joint Publication of ISTE & AVIXA

AV Fundamentals for Educators

Lab Workbook

Companion to the AV Fundamentals for Educators Textbook • First Edition

12 Structured Laboratory Exercises for Classroom Educators

Designed for K-12 and Higher Education Instructors • Consumer-to-Prosumer Equipment • Scalable for All Budgets

International Society for Technology in Education (ISTE) • Arlington, VA
AVIXA — Audiovisual and Integrated Experience Association • Fairfax, VA

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"Empowering Educators to Understand, Evaluate, and Advocate for Effective AV in Learning Spaces."

How to Use This Workbook

Purpose and Audience

This workbook is designed for **classroom educators, instructional technology specialists, and education administrators** who are enrolled in or facilitating the *AV Fundamentals for Educators* course. It is **not** designed for AV installation

technicians or systems integrators. Each laboratory exercise emphasizes the educator's role: understanding, evaluating, planning for, and advocating on behalf of effective AV systems in learning environments.

Structure of Each Lab

Each of the 12 labs in this workbook corresponds directly to a chapter in the textbook. Labs follow a consistent structure:

1. **Objective** — What you will accomplish and learn
2. **Materials** — Equipment and software required (with budget-flexible alternatives)
3. **Safety Reminders** — Lab-specific safety notes (always supplement with the Universal Safety Checklist, p. 7)
4. **Pre-Lab Preparation** — Reading, vocabulary review, and prior knowledge activation
5. **Procedure** — Numbered, step-by-step instructions
6. **Worksheet(s)** — Fillable tables, diagrams, and measurement logs
7. **Reflection Questions** — Critical thinking questions for written or discussion response
8. **Rubric Reference** — Points to the corresponding rubric in the Instructor Guide

Completing Your Work

- Read the corresponding textbook chapter **before** arriving at lab.
- Complete the Pre-Lab section independently or with your lab partner before lab begins.
- Complete the Universal Safety Checklist (page 7) at the start of every lab session.
- Record all measurements, observations, and findings **directly in the worksheet tables**. Accuracy and completeness are valued over perfection.

- Submit reflection questions in the format specified by your instructor (written, digital, or discussion board).
- If equipment is unavailable, refer to **Appendix E: Alternative Lab Options** for simulation and observation alternatives.

A Note on Equipment

Labs in this workbook are designed to be **scalable**. Where possible, materials lists include free software alternatives, smartphone apps, and consumer-grade substitutes. If your institution lacks specific equipment, consult Appendix E for virtual simulation links, partner institution options, and YouTube-based observation alternatives. No lab requires proprietary or enterprise-only tools.

Working with Partners

Most labs can be completed individually or in pairs. Where a second person is needed (e.g., remote participant in Lab 7), your instructor will coordinate pairing or use a recording-based alternative. Communication, documentation, and observation are the core skills — not hardware proficiency.

Tip: Digital vs. Paper Worksheets

Your instructor may provide digital versions of these worksheets (fillable PDF or editable document). If completing on paper, write legibly in ink and retain originals. Submit copies if instructed to do so per the Lab Report Submission Guidelines on page 9.

Safety Overview: General AV Lab

Safety Rules

SAFETY FIRST — Read Before Any Lab Activity

AV equipment operates on standard AC power and generates heat. Improper handling can result in electrical shock, hearing damage, equipment damage, or trip hazards. All participants must review this safety overview before the first lab session and whenever starting a new lab.

Electrical Safety

- Inspect all power cables for cuts, fraying, or damaged insulation before plugging in.
- Never remove equipment covers or attempt to repair internal components. Report malfunctioning equipment to your instructor immediately.
- Use only properly grounded three-prong outlets. Do not use adapters to defeat ground pins.
- Never operate equipment with wet hands or near liquids of any kind.
- Power off equipment before connecting or disconnecting signal cables to reduce transient voltage spikes.
- Do not daisy-chain power strips. Use surge-protected power distribution units (PDUs) rated for the load.
- Allow projectors and amplifiers to cool down for at least 10 minutes before storing or transporting.

Cable Management

- Route all cables along walls or under cable cover ramps — never across walkways.
- Label both ends of every cable you deploy using provided cable labels.
- Do not bend cables beyond their minimum bend radius; coil excess cable loosely (the "over-under" technique for XLR and audio cables).
- Remove all temporary cable runs at the end of each lab session. Return cables to their storage bags or reels.
- Report any cable damage (fraying, bent pins, cracked connectors) to the instructor immediately — do not use damaged cables.

Equipment Handling

- Lift equipment using proper technique: bend knees, keep back straight, and ask for assistance with items over 30 lbs.
- Never place drinks, food, or any liquids on or near AV equipment.
- Handle display screens and camera lenses with clean, dry hands. Do not touch lens surfaces directly.
- Microphones are fragile precision instruments — do not drop, tap, or blow forcefully into dynamic microphones.
- Amplifiers and powered speakers: always set volume to zero before powering on; bring up volume gradually.
- Laser projectors: never look directly into the beam aperture, even when the projector appears to be off.

Hearing Safety

- Sound pressure levels (SPL) should not exceed **85 dB SPL** during any lab exercise.
- Hearing protection (foam earplugs) is available in the lab supply cabinet.

- If intentionally inducing audio feedback (Lab 2), do so at minimal volume and for the shortest time necessary.

Emergency Procedures

- The main power disconnect for the AV lab is located:
- In the event of electrical fire, use a Class C fire extinguisher — location:
- Emergency exit is located:
- Campus emergency number:

ADA Workspace Requirement

All lab workstations must be accessible to participants using wheelchairs or other mobility aids. Minimum aisle width: 36 inches. Equipment controls must be reachable from a seated position. If a workstation is not accessible, notify the instructor before the lab begins so alternative arrangements can be made. See the Accessibility Statement (page 10) for full accommodation details.

Lab Equipment Inventory Checklist

*Master inventory of recommended equipment for all 12 labs. Items marked * have free or low-cost alternatives. See Appendix E for substitution options.*

Category	Item Description	Qty Recommended	Used in Labs	Alternative / Note	In Stock ✓
Display	LCD/LED	1-2	3, 7, 9, 10	Consumer	

Category	Item Description	Qty Recommended	Used in Labs	Alternative / Note	In Stock ✓
	Flat Panel Display (32" or larger)			TV acceptable	
Display	Data Projector (short-throw or standard)	1	3, 8, 10	Any lumen class	
Audio	Dynamic Microphone (e.g., Shure SM58 or equivalent) *	2	2, 7	Any USB mic as alternative	
Audio	Condenser Microphone (small diaphragm) *	1	2	Optional; compare to dynamic	
Audio	Audio Interface or Small Mixer (2-4 channel)*	1	2, 7	USB audio interface acceptable	
Audio	Powered Monitor Speaker (5" or larger)*	2	2, 7, 10	Computer speaker as substitute	
Cables	XLR Cables (10 ft) *	4	2, 4, 10		
Cables	TRS/TS Audio Cables (various)*	4	2, 4		
Cables	HDMI Cables (6 ft)*	6	3, 4, 7, 10		
Cables	USB-A, USB-C, DisplayPort cables	4 ea.	4, 5		

Category	Item Description	Qty Recommended	Used in Labs	Alternative / Note	In Stock ✓
	(assorted)				
Cables	Cat5e/Cat6 Ethernet Patch Cables (6 ft)*	8	5, 10		
Cables	VGA Cable (legacy reference only)*	1	4		
Networking	Managed Network Switch (8-16 port)*	1	5	Unmanaged switch for budget	
Networking	Wireless Access Point (lab-isolated network)	1	5, 7		
Control	AV Control System Touchpanel (physical or simulated)*	1	6	Crestron, Extron, Q-SYS, or tablet simulation	
Video Conf.	Webcam (wide-angle, 1080p minimum)*	2	7	Built-in laptop camera acceptable	
Computers	Laptop or Desktop Computer*	1 per student	All labs	Chromebook for most labs	
Computers	Smartphone (for measurement apps)*	1 per student	1, 2, 3		
Measurement	Tape Measure (25 ft)*	2	1, 3, 8		
Measurement	Multimeter (basic)*	1	10	Optional; borrow	

Category	Item Description	Qty Recommended	Used in Labs	Alternative / Note	In Stock ✓
				from school	
Accessories	Cable Label Kit*	1 set	4, 5	Masking tape + marker acceptable	
Accessories	Clipboard and graph paper	1 per student	1, 8		
Safety	Foam Earplugs (box of 50 pairs)	1 box	2		
Safety	Cable Cover Ramps / Floor Cord Covers	4	All powered labs		
Safety	Surge-Protected PDU (6–8 outlet)*	2	All powered labs		
Reference	Cable/ Connector Identification Kit (sample connectors)	1 set	4, App. B		
Software (Free)	Audacity (audio recording/editing)*	—	2	Browser-based alternatives exist	
Software (Free)	Angry IP Scanner or equivalent*	—	5		
Software (Free)	draw.io (diagramming)*	—	1, 4, 5	Paper diagram acceptable	
Software (Free)	NIOSH SLM App (SPL measurement)*	—	2	Any SPL meter app	

Category	Item Description	Qty Recommended	Used in Labs	Alternative / Note	In Stock ✓
Software (Free)	DisplayCAL or built-in test patterns*	—	3		
Software (Free)	WebAIM Contrast Checker*	—	6, 9	browser-based	
Software (Free)	Zoom, Teams, or Meet (video conferencing)*	—	7		
Software (Free)	Canva, Google Slides, or PowerPoint*	—	9		

Instructor: Complete the "In Stock" column and attach procurement requests for missing items to this checklist. Update at the start of each semester.

Lab Report Submission Guidelines

What to Submit

Each completed lab produces a **Lab Report Package** consisting of:

9. Completed Universal Safety Checklist (signed and dated)
10. Completed Worksheet(s) — all tables filled in, diagrams drawn or attached
11. Completed Reflection Questions (written responses, minimum 3 sentences each)

12. Any photographs or screenshots required by the procedure (labeled with lab number and step)
13. Instructor-specified additional artifacts (e.g., signal flow diagram, presentation file, memo)

Format Requirements

Item	Requirement
Worksheets (paper)	Original paper worksheets, legibly completed in ink
Worksheets (digital)	Fillable PDF or editable document; file name: LastName_Lab#_Worksheet.pdf
Reflection Questions	Typed responses preferred; minimum 3 complete sentences per question
Photographs	JPG or PNG; labeled; embedded in document or submitted as separate folder
Diagrams	Hand-drawn (photographed clearly) or digital (draw.io export, PDF preferred)
Presentations (Labs 9, 12)	PowerPoint, Google Slides, or PDF; submitted before presentation date

Submission Deadlines

Lab Report Packages are due **one week after the lab session** unless otherwise noted by the instructor. Labs involving presentations (Labs 9, 12) have two deliverables: the presentation file (due before class) and the written summary (due one week after presentation).

Academic Integrity

Collaboration on lab procedures is encouraged. Worksheets and reflection questions must represent your own observations and written responses. Direct copying of a partner's worksheet entries or reflection responses is an academic integrity

violation. Measured values (SPL, lux, dimensions) may differ between partners if recorded independently at different positions — this is expected and acceptable.

Grading Summary

Component	Points (Typical)	Notes
Safety Checklist	5	Must be complete to earn credit for lab
Completed Worksheets	40	Accuracy + completeness
Reflection Questions	35	Depth of analysis and critical thinking
Additional Artifacts	20	Diagrams, photos, presentations
Total	100	Per lab; see Instructor Guide for rubrics

Accessibility Statement

ISTE and AVIXA are committed to providing equitable access to all course materials and laboratory experiences. This workbook and all associated activities are designed in alignment with the **Americans with Disabilities Act (ADA), Section 508** of the Rehabilitation Act, and **WCAG 2.1 Level AA** guidelines where applicable.

Physical Lab Accessibility

- All lab workstations must maintain a minimum aisle width of 36 inches (91 cm) for wheelchair access.
- Equipment controls should be reachable from a seated position (maximum height 48 inches per ADA standards).

- Lab tables should be height-adjustable or have knee clearance of at least 27 inches.
- If a standard workstation is not accessible, the instructor will provide an equivalent modified station before the lab begins.

Document Accessibility

- Digital versions of all worksheets are available as accessible fillable PDFs with proper reading order, tagged headings, and alt text on all images.
- Large-print versions (18pt or larger) of all worksheets are available upon request.
- All electronic documents use a minimum color contrast ratio of 4.5:1 (WCAG AA) for body text.

Accommodation Requests

Students requiring accommodations should contact their instructor or institution's accessibility services office **at least one week before** the first lab session. Typical accommodations available include:

- Extended time for lab procedures and written reflections
- Partner or group completion of physical measurement tasks
- Alternative lab formats (virtual simulation, video observation) — see Appendix E
- Captioning or sign language interpretation during lab demonstrations
- Screen reader-compatible digital worksheets
- Reduction of sensory stimuli (dimmed lights, reduced audio volume) upon request

Alternative Lab Options

Every lab in this workbook has at least one **equipment-free or remote alternative**. These alternatives are designed to achieve equivalent learning

outcomes using free online tools, video observation, or virtual simulation. See **Appendix E** for the complete Alternative Lab Options guide. Instructors are encouraged to pre-approve alternative formats for any student or school with equipment or facility limitations.

Hearing Safety in AV Labs

Labs involving amplified audio (Labs 2, 7, 10) will keep SPL at or below 85 dB. Foam hearing protection is always available. Students with hearing sensitivity or hearing loss should inform the instructor so that visual monitoring aids (level meters displayed on screen) can be used throughout the session.

Accessibility as Curriculum

Several labs in this workbook — particularly Labs 6, 7, and 11 — directly assess the accessibility of AV systems. Students are encouraged to bring their own perspective and lived experience to these evaluations. There are no wrong answers when documenting genuine usability observations.

Universal Safety Checklist

Complete at the beginning of every lab session. Retain with your Lab Report Package.

AV Lab Safety Checklist — Required Before Every Lab

Student Name:

Lab Number & Title:

Date: **Instructor Present:**

Place a checkmark in each box after verifying the item. Initial and date at the bottom.

✓ / ✗

Safety Item

Notes / Corrective Action Taken

All power cables and signal cables inspected for visible damage (cuts, fraying, exposed wires, bent pins)

All powered equipment properly grounded (three-prong plugs fully seated in grounded outlets)

No liquids (drinks, water bottles) present within 3 feet of any powered AV equipment

Emergency stop / main power disconnect location reviewed and understood

Instructor (or authorized lab monitor) present and aware that power-on is imminent

Personal protective equipment (hearing protection) available and location known

ADA accessibility of workspace verified (36-inch aisles clear, controls reachable)

All cable runs routed away from walkways or covered with cord protectors

Volume / output levels set to minimum (zero) before powering on amplified equipment

Emergency exit location identified and path is unobstructed

Fire extinguisher location identified (Class C for electrical fires)

Equipment power ratings checked — not exceeding PDU or outlet capacity

Student Initials: **Date:**

Instructor Verification Initials: **Time Lab Began:**

Note: Reproducible Page

This checklist may be photocopied or printed for each lab session. One completed checklist is required per student per lab. Checklists must be attached to the Lab Report Package at submission.

LAB 1: Exploring the AV Ecosystem

Corresponding to Chapter 1 • Estimated Time: 75 Minutes • Difficulty: Foundational

Objective

Map and document the AV components of an existing classroom or learning space. Students will identify and categorize every AV element in the room, assess accessibility features, and produce an initial signal flow diagram and written recommendations.

Materials

- Clipboard and pencil or pen
- Measuring tape (25 ft / 7.6 m)
- Smartphone or camera (photograph capability)
- AV Inventory Worksheet (this lab, page below)
- Free diagramming app: draw.io (app.diagrams.net) — or pencil and graph paper

Safety Reminder

Complete the Universal Safety Checklist before this lab. Do not power on, unplug, or move any installed AV equipment. This is an observation-only lab. Do not climb on furniture to inspect ceiling-mounted equipment — observe and document from the floor.

Pre-Lab Preparation

- Review Chapter 1 vocabulary, especially: *source, processing, distribution, output, signal flow, AV system, audiovisual*.
- Sketch a rough floor plan of your lab room from memory before arriving.
- Identify one AV system you interact with daily and describe it in one sentence in the space below.

Pre-Lab Response

AV system I interact with daily (describe in one sentence):

Procedure

Step 1 — Identify Displays: Locate all display devices in the room (projectors, flat panels, monitors, digital signage, interactive whiteboards). Record each in the inventory table.

Step 2 — Identify Audio Equipment: Locate all audio components (speakers, microphones, amplifiers, mixers, hearing loop components). Note whether speakers are ceiling, wall, or floor-standing mounted.

Step 3 — Identify Control Systems: Find the room's AV control interface. This may be a wall panel, touchpanel, remote control, or educator station PC. Document its location and any visible brand or model information.

Step 4 — Identify Connectivity Points: Locate all input panels, wall plates, and connection ports available in the room (HDMI, USB, VGA, audio jacks, network jacks, wireless presentation systems). Document their location and types.

Step 5 — Document Cable Types: Observe visible cables. Identify cable types you can see (HDMI, XLR, speaker wire, Cat6, power). Note any cables that are unlabeled, exposed, or appear damaged — do NOT touch or move them, only observe.

Step 6 — Photograph the System: Take at least 5 photographs: (a) overall room from the back, (b) display(s), (c) audio equipment, (d) control interface, (e) a connectivity panel or wall plate. Label photos with the room name and component photographed.

Step 7 — Complete the Inventory Table: Transfer all observations to the AV Room Inventory Table below. Fill in all columns for every component identified.

Step 8 — Draw a Signal Flow Diagram: Using the Signal Flow Template below, draw arrows connecting identified components from Source → Processing → Distribution → Output. You do not need to know every connection — document what you can trace or reasonably infer.

Step 9 — Note Accessibility Features: For each component, note whether it has an identifiable accessibility feature (caption output, hearing loop transmitter, high-contrast labeling, audio description output, etc.).

Step 10 — Write Three Recommendations: Based on your observations, write three specific, actionable recommendations for improving this AV system in the Reflection section.

Worksheet: AV Room Inventory Table



Room Name / Location: _____ **Date:** _____

Component Type	Brand / Model (if labeled)	Location in Room	Condition (Good / Fair / Poor)	Accessibility Feature? (Y / N / Unknown)	Notes
Display (Projector)					
Display (Flat Panel)					
Display (Other)					
Speaker (L)					
Speaker (R)					
Speaker (Sub / Center)					
Microphone					
Amplifier / Receiver					
AV Switcher / Scaler					
Control Interface					
Source (Computer / PC)					
Source (Blu-ray / Media Player)					
Input Panel / Wall Plate					
Wireless Presentation System					
Video Conferencing System					

Component Type	Brand / Model (if labeled)	Location in Room	Condition (Good / Fair / Poor)	Accessibility Feature? (Y / N / Unknown)	Notes
Digital Signage Player					
Hearing Loop / Assistive Listening					
Other: _____					
Other: _____					

Signal Flow Diagram Template

 Draw arrows connecting boxes to show signal path. Add component names inside each box.



Add additional boxes below for complex systems. Arrows may branch (one source to multiple outputs).

Reflection Questions



Rubric Reference:

See Lab 1 Rubric in the Instructor Guide. Key dimensions: Completeness of Inventory Table (40%), Signal Flow Diagram accuracy (25%), Reflection depth (35%).

LAB 2: Audio Signal Chain Exploration

Corresponding to Chapter 2 • Estimated Time: 90 Minutes • Difficulty: Foundational–Intermediate

Objective

Trace and test a complete audio signal chain from microphone to speaker. Measure sound pressure levels (SPL) at defined positions, observe the effect of EQ adjustments, intentionally introduce and correct audio feedback, and document all findings in a structured log.

Materials

- Dynamic microphone (e.g., Shure SM58 or equivalent) — XLR type preferred
- Condenser microphone (optional, for polar pattern comparison)
- Small audio mixer or USB audio interface (2–4 channel)
- Powered monitor speaker or computer speaker
- XLR cables (2), TRS cables (1)
- Laptop with Audacity (free: audacityteam.org) installed
- NIOSH SLM app or equivalent SPL meter app on smartphone
- Measuring tape

Hearing Safety — Critical for This Lab

Always start mixer gain at minimum. Bring volume up gradually. Keep SPL below 85 dB at all times. Foam hearing protection must be available and accessible throughout this lab. When introducing intentional feedback in Step 8, do so ONLY

at the lowest volume that produces audible feedback — reduce volume immediately. Duration of feedback should not exceed 3 seconds.

Pre-Lab Preparation

- Review Chapter 2 vocabulary: *frequency, amplitude, decibel (dB), SPL, dynamic range, gain, EQ (equalization), feedback, polar pattern (cardioid, omnidirectional)*.
- Download and install (or browser-test) Audacity before the lab session.
- Install the NIOSH SLM app (free, iOS/Android) on your smartphone.

Procedure

Step 1 — Connect Microphone to Mixer: Using an XLR cable, connect the dynamic microphone to Channel 1 input of the mixer. Ensure all faders and gain controls are set to minimum before connection.

Step 2 — Set Initial Gain: With the microphone connected, speak into it from approximately 6 inches away at a normal speaking voice. Slowly increase the Channel 1 gain until the level meter reads approximately -12 dBVU (just below 0). This is your working gain level. Record it in the worksheet.

Step 3 — Connect Mixer to Speaker: Using a TRS or XLR cable, connect the mixer's main output (L or Mono) to the powered monitor speaker input. Ensure the speaker's volume control is at minimum before this connection.

Step 4 — Bring Up Speaker Volume and Test: Slowly increase the speaker volume to a comfortable listening level. Speak into the microphone. Confirm audio is passing through to the speaker. Adjust levels as needed — signal should be clear, not distorted.

Step 5 — Baseline SPL Measurement: Position the smartphone with SPL meter app at 1 meter from the speaker. Speak continuously into the microphone at your normal teaching voice level for 10 seconds. Record the measured SPL in the worksheet. Repeat at 2 meters and 3 meters.

Step 6 — Record Levels: Open Audacity. Set it to monitor input from the audio interface or microphone. Observe the waveform while speaking at normal volume. Take a screenshot or note the peak dBFS level displayed. Record in the worksheet.

Step 7 — EQ Observation: Locate the 3-band or parametric EQ on your mixer channel. Boost the LOW band (+6 dB, approximately 80–150 Hz) and speak into the mic. Note what you hear. Return to flat. Boost the HIGH band (+6 dB) and note what you hear. Return to flat. Record observations in the EQ Comparison Table.

Step 8 — Intentional Feedback (Low Level): With the instructor present, slowly move the microphone toward the speaker while speaking. At the first sign of feedback (ringing or howling), immediately pull the microphone back or reduce channel gain. Do NOT allow feedback to persist. Record the approximate microphone-to-speaker distance at which feedback began. This is the "feedback threshold" for this setup.

Step 9 — Polar Pattern Comparison (if condenser available): Replace the dynamic mic with the condenser mic. Set the condenser to cardioid. Measure SPL at 1 meter. Then set to omnidirectional (if available). Note any difference in level or feedback sensitivity. Record in the worksheet.

Step 10 — Document Full Signal Chain: Draw the complete signal chain for this lab in the diagram space below. Label every component, cable type, and connector type at each junction.

Step 11 — Disconnect and Store: Set all volumes to zero. Power down speaker, then mixer, then disconnect cables in order from output to source. Coil cables using the over-under technique and return to storage.

Step 12 — Create Labeled Sketch: In the diagram space below, create a neatly labeled sketch of your physical setup showing approximate distances and component positions.

Worksheet: Audio Signal Chain Documentation



Audio Signal Path — Fill in each segment

Segment #	From Component	To Component	Cable Type	Connector A	Connector B	Signal Type
1	Microphone			XLR F		Mic-level audio
2						
3						
4		Speaker				Line-level audio

SPL Measurement Log

Measurement #	Distance from Speaker	Measured dB SPL	Speaking Level	Notes
1	1 meter		Normal speaking voice	
2	2 meters		Normal speaking voice	
3	3 meters		Normal speaking voice	
4	1 meter		Loud/projected voice	
5	1 meter		Quiet/whisper	

EQ Comparison Table

EQ Setting	Frequency Band Adjusted	Amount (+/- dB)	Observed Tonal Character	Appropriate Use Case?
Flat (baseline)	None	0 dB		
Low Boost	~80-150 Hz	+6 dB		
High Boost	~8-12 kHz	+6 dB		

EQ Setting	Frequency Band Adjusted	Amount (+/- dB)	Observed Tonal Character	Appropriate Use Case?
Custom Setting				

 Feedback Log

Item	Recorded Value
Approximate distance at which feedback began	
Channel gain setting at time of feedback	
Corrective action taken (describe)	
Estimated time before feedback was corrected (seconds)	

 Signal Chain Sketch (draw below)

Reflection Questions



Rubric Reference:

See Lab 2 Rubric in the Instructor Guide. Key dimensions: SPL measurement accuracy and log completeness (35%), EQ and feedback documentation (30%), Reflection analysis (35%).

LAB 3: Video Display Evaluation

Corresponding to Chapter 3 • Estimated Time: 80 Minutes • Difficulty: Foundational-Intermediate

Objective

Compare and evaluate two or more display technologies (projector vs. flat panel preferred) using visual assessment, light measurement, and standardized test patterns. Assess each display's suitability for classroom use, including accessibility considerations.

Materials

- Two display devices (projector + flat panel display preferred)
- Laptop computer with HDMI output
- Test pattern source: DisplayCAL (displaycal.net), built-in OS test patterns, or downloaded test card images
- Lux meter app (smartphone — e.g., Lux Light Meter Pro or equivalent)
- Measuring tape
- Ruler or tape for screen size measurement



Projector Safety

Never look directly into the projector lens or beam aperture, even when the projector appears off. Allow projector lamps to cool for at least 10 minutes before handling or moving. Lamp-based projectors reach very high temperatures internally.

Pre-Lab Preparation

- Review Chapter 3 vocabulary: *resolution, lux, lumens (ANSI lumens), contrast ratio, throw ratio, viewing angle, color gamut, refresh rate, pixel pitch*.
- Download and open test pattern images on your laptop before lab begins.
- Measure the room's ambient light level (lux) with the room lights on and record it below:

Pre-Lab: Ambient Light Measurement

Measured ambient light (room lights ON): _____ lux Measured ambient light (room lights OFF): _____ lux

Procedure

Step 1 — Display Test Pattern: Connect the laptop to Display A. Display a full-screen test pattern (grayscale ramp, color bars, and a fine detail resolution chart). Allow the display to warm up for 5 minutes before measurement.

Step 2 — Measure Ambient Light: Measure room ambient light with lux meter app at the center of the audience seating area. Record with room lights on and off.

Step 3 — Evaluate Brightness (Display A): Using the lux meter, measure the light output at the center of the screen surface from 1 meter away. Record in the comparison matrix. Visually assess overall brightness as Bright / Adequate / Dim for the current room lighting condition.

Step 4 — Evaluate Contrast Visually: Display an alternating black-and-white checkerboard test pattern. Observe whether black areas appear truly dark or washed out. Rate contrast as Excellent / Good / Fair / Poor.

Step 5 — Measure Viewing Angle: Walk slowly from the center-front of the display to the far left and far right, observing when the image quality visibly degrades (washes out or color shifts). Estimate the maximum useful viewing angle in degrees from center. Record for each display.

Step 6 — Measure Throw Ratio (Projector only): Measure the distance from the projector lens to the screen surface (throw distance). Measure the width of the projected image on screen. Calculate: $\text{Throw Ratio} = \text{Throw Distance} \div \text{Screen Width}$. Record in the Throw Ratio Calculation template.

Step 7 — Check Color Accuracy: Display a standard color bar test pattern (SMPTE bars or similar). Visually assess whether red, green, blue, cyan, magenta, and yellow bars appear accurate (not shifted, faded, or oversaturated).

Step 8 — Test Resolution: Display a 1080p resolution test chart. Count the number of distinct lines visible in the center resolution wedge. Note whether the display upscales or natively renders 1080p content.

Step 9 — Test Caption Visibility: Display a slide with white closed-caption-style text on black bar at the bottom of the screen (minimum 28pt text). Assess legibility from the back of the room for each display.

Step 10 — Repeat for Display B and Document: Repeat Steps 1–9 for Display B. Complete the full comparison matrix and make your written recommendation.

Worksheet: Display Comparison Matrix

Display Evaluation Results

Criterion	Display A (Projector)	Display B (Flat Panel)	Advantage
Native Resolution			
Measured Brightness (lux at 1m)			
Brightness Visual Rating			
Contrast (visual assessment)			
Max Useful Viewing Angle (degrees)			
Color Accuracy (visual)			
Throw Ratio (projector only)		N/A	N/A
Caption Text Legibility (back of room)			
Accessibility Features Noted			
Cost Category (Low / Med / High)			
Ease of Use for Educator			
Overall Recommendation			

Throw Ratio Calculation (Projector)

Measurement	Value	Unit
Throw Distance (lens to screen)		feet / inches
Projected Image Width on Screen		feet / inches
Throw Ratio = Distance ÷ Width		ratio (e.g., 1.5:1)
Projector Specification Throw Ratio (label/manual)		ratio
Match? (Yes / No / Within 10%)		—

Reflection Questions



Rubric Reference:

See Lab 3 Rubric in the Instructor Guide. Key dimensions: Measurement accuracy and comparison matrix completeness (40%), Throw ratio calculation (20%), Reflection quality (40%).

LAB 4: Signal Flow Tracing and Documentation

Corresponding to Chapter 4 • Estimated Time: 80 Minutes • Difficulty: Intermediate

Objective

Physically trace and document signal flow in an existing AV system. Produce a complete, accurate signal flow diagram with annotated cable types, signal types, and active/passive component identification.

Materials

- Existing AV system in a classroom or AV demo room
- Pencil and graph paper (or laptop with draw.io)
- Cable identifier labels and permanent marker
- Flashlight or phone flashlight (to inspect cable runs behind equipment)

- Camera or smartphone for photos



Safety Reminder

Do not disconnect or move any installed cables during this lab without instructor permission. Disconnecting an installed AV system without authorization may disrupt active classroom use elsewhere in the building. Observe only; do not modify the system.

Pre-Lab Preparation

- Review Chapter 4 vocabulary: *signal flow, passive component, active component, switcher, scaler, extender, balun, HDBaseT, signal integrity, impedance, analog signal, digital signal*.
- Bring the completed Lab 1 inventory for the same room if available — it will serve as a starting reference.

Procedure

Step 1: List all source devices identified in the room. Write them in Column A of the Documentation Table.

Step 2: List all output devices in the room. Write them in the table.

Step 3: Physically trace each cable run — follow from one end to the other. Where cables run through walls or ceiling, note entry and exit points.

Step 4: Document the cable type (HDMI, Cat6 with balun, XLR, coax, etc.) and estimated length for each segment.

Step 5: Identify any active components in the signal path (switchers, scalers, amplifiers, HDBaseT transmitters/receivers). Note brand/model if labeled.

Step 6: Create a draft signal flow diagram using the 10×10 grid on the worksheet. Place sources on the left, outputs on the right.

Step 7: Follow the signal from each source to the output by walking the cable run. Verify your diagram matches the physical installation.

Step 8: Add annotations to your diagram: label each cable segment with cable type and length; label active components; mark any signal conversion points.

Step 9: Identify any observable signal integrity issues (e.g., bent cables, unlabeled connections, frayed connectors, missing strain relief). Note in the table.

Step 10: Finalize and clean up your diagram. Compare with a partner's diagram — note any discrepancies and reconcile.

Worksheet: Signal Flow Documentation Table

AV System Signal Flow Documentation

Seg #	Source Component	Destination Component	Cable Type	Length (ft)	Connector A	Connector B	Signal Type (A/V/Data)	Active? (Y/N)	Notes / Issues
1									
2									
3									
4									
5									
6									
7									
8									

 Signal Flow Diagram Grid (10×10) — Use boxes for components, arrows for signal direction

Reflection Questions



Rubric Reference:

See Lab 4 Rubric in the Instructor Guide. Key dimensions: Signal flow diagram completeness and accuracy (50%), Documentation table (25%), Reflection quality (25%).

LAB 5: Network Discovery for AV

Corresponding to Chapter 5 • Estimated Time: 75 Minutes • Difficulty: Intermediate

Objective

Identify and document AV-related devices on a designated lab network. Practice IP configuration documentation, web interface access, and basic network topology mapping in the context of networked AV systems.

Materials

- Computer with network access (connected to designated lab subnet only)
- Angry IP Scanner (free: angryip.org) or equivalent (Advanced IP Scanner, Fing)
- Web browser
- Managed switch (optional, for VLAN discussion)
- Network patch cables

 **Critical: Network Ethics and Permission**

Perform ALL network scanning activities ONLY on the designated, isolated lab subnet provided by your instructor. Never scan the school's production network, the internet, or any network segment not explicitly designated for this lab. Unauthorized network scanning may violate school policy, state law, and federal law (Computer Fraud and Abuse Act). If you are unsure whether you are on the correct network, stop and ask your instructor before proceeding.

Pre-Lab Preparation

- Review Chapter 5 vocabulary: *IP address, MAC address, subnet, DHCP, DNS, VLAN, QoS, bandwidth, latency, OUI (Organizationally Unique Identifier), firmware, web interface.*
- Your instructor will provide the lab subnet address (e.g., 192.168.10.0/24) before the session begins.

Pre-Lab: Designated Lab Subnet (provided by instructor)

Lab Subnet: _____ Gateway: _____ DHCP Range:

My Computer's IP Address: _____ My Computer's MAC:

Procedure

Step 1: Launch Angry IP Scanner. Confirm the scanning range matches the designated lab subnet. Do not modify the range without instructor approval.

Step 2: Run the scan. Wait for it to complete fully before recording results.

Step 3: Review scan results. For each device with a "Alive" status, record its IP address and hostname (if resolved). Note any devices showing as offline or timed out.

[illegible]

Device Name / Host name	IP Address	MAC Address	Manufacturer (OUI)	Firmware Version	Web Interface (Y/N)	Link Speed	Avg Ping (ms)	Packet Loss %	VLAN (if set)	Notes

 QoS Discussion Notes

Traffic Type	Priority Level (High/Med/Low)	Reason	Recommended DSCP Tag (if known)
Video Conferencing (live)			
Audio Streaming			
AV Control Signals			
Digital Signage Content			
Firmware Updates			

 **Network Topology Diagram (draw below — include switch, all devices, IP addresses)**

Reflection Questions



Rubric Reference:

See Lab 5 Rubric in the Instructor Guide. Key dimensions: Network inventory completeness (40%), Topology diagram (20%), QoS analysis (15%), Reflection quality (25%).

LAB 6: Control System Interface Evaluation

Corresponding to Chapter 6 • Estimated Time: 80 Minutes • Difficulty: Intermediate

Objective

Evaluate an AV control system touchpanel interface for usability and accessibility. Apply Nielsen's heuristic evaluation methodology, assess WCAG touch target compliance, and produce a prioritized list of improvement recommendations.

Materials

- AV control system touchpanel (Crestron, Extron, Q-SYS, or equivalent) — physical or software simulation
- WebAIM Contrast Checker (webaim.org/resources/contrastchecker/) — browser-based, free
- Ruler or on-screen measurement tool
- Camera or smartphone for screenshots
- Heuristic Evaluation Form (this worksheet)

About Heuristic Evaluation

A heuristic evaluation is a usability inspection method in which evaluators examine an interface and judge its compliance with recognized usability principles (heuristics). Originally defined by Jakob Nielsen (1994), the 10 heuristics below are widely used in UX design. You are applying them here to AV control system interfaces — a practical and educationally relevant domain.

Pre-Lab Preparation

- Review Chapter 6 vocabulary: *control system, touchpanel, UI (user interface), heuristic, WCAG, touch target, color contrast, feedback (in UI context), cognitive load*.
- Read the WCAG 2.1 Success Criterion 2.5.5 (Target Size): interactive targets should be at least 44×44 CSS pixels (approximately 44×44pt at 96 dpi).
- Photograph or screenshot the touchpanel UI before beginning the evaluation.

Procedure

Step 1: Document the room the control system serves (room name, approximate size, primary use).

Step 2: Photograph or screenshot the entire touchpanel UI. Attach to your Lab Report Package.

Step 3: Identify and list all control functions available on the touchpanel (power, source select, volume, display, camera, lighting, etc.).

Step 4: Test each control function by pressing it. Confirm it produces the expected result. Note any functions that do not work as expected.

Step 5: Measure or estimate the size of the 5 smallest interactive buttons. Compare to the 44×44pt WCAG minimum. Record in the accessibility checklist.

Step 6: Using WebAIM Contrast Checker, enter the hex color codes for button text and button background color. Record the contrast ratio. WCAG AA requires 4.5:1 for normal text, 3:1 for large text (18pt+ or 14pt bold).

Step 7: Test audio/haptic feedback: does the panel provide tactile or audio confirmation when a button is pressed? Rate as Present / Absent.

Step 8: Evaluate navigation flow: how many taps does it take to complete the most common educator task (e.g., "power on projector, select laptop input, set volume to 50%")? Count and record.

Step 9: Identify your top 3 usability improvements. For each, state the heuristic violated, the specific problem, and your proposed solution.

Step 10: Write a brief (3-5 sentence) accessibility evaluation summary for this control system interface.

Worksheet: Control System Heuristic Evaluation

 **Heuristic Evaluation Rating Scale: 0 = No problem | 1 = Cosmetic | 2 = Minor | 3 = Major | 4 = Catastrophic**

#	Heuristic	Rating (0-4)	Specific Problem Found (or "None")	Severity Justification
1	Visibility of System Status — does the UI show current state clearly? (e.g., which source is active, current volume level)			
2	Match Between System and Real World — does the UI use language and icons familiar to educators (not AV jargon)?			
3	User Control and Freedom — can the user undo mistakes easily? Is there an "all off" or "help" button?			
4	Consistency and Standards — are similar			

#	Heuristic	Rating (0-4)	Specific Problem Found (or "None")	Severity Justification
	functions labeled and styled consistently throughout the interface?			
5	Error Prevention — does the UI prevent common mistakes (e.g., accidental system power off without confirmation)?			
6	Recognition Rather Than Recall — can users identify functions from the screen without memorization ?			
7	Flexibility and Efficiency — can experienced users complete tasks quickly? Are shortcuts available?			
8	Aesthetic and Minimalist Design — is the UI free of irrelevant or rarely needed information?			
9	Help Users Recognize, Diagnose, and			

#	Heuristic	Rating (0-4)	Specific Problem Found (or "None")	Severity Justification
	Recover from Errors — are error messages clear and actionable?			
10	Help and Documentation — is help available on the panel or nearby? Is it accessible?			

Accessibility Checklist for Touch Interface

Item	Status (Pass / Fail / N/A)	Notes
Smallest button $\geq$ 44×44pt in size		
Button text contrast ratio $\geq$ 4.5:1 (measured with WebAIM)		
Touch feedback (tactile or auditory) present		
No information conveyed by color alone		
Emergency / all-off function clearly visible and accessible		
Interface usable with one hand / limited dexterity		
Font size legible from standing operating position		
Panel height accessible from wheelchair position ($\leq$ 48 in from floor)		

UI Improvement Recommendations

#	Heuristic Violated	Problem Description	Proposed Solution	Est. Effort (Low/Med/High)
1				
2				
3				

Reflection Questions



Rubric Reference:

See Lab 6 Rubric in the Instructor Guide. Key dimensions: Heuristic evaluation rigor (35%), Accessibility checklist (25%), Recommendations quality (20%), Reflection (20%).

LAB 7: Video Conferencing Room Audit

Corresponding to Chapter 7 • Estimated Time: 90 Minutes • Difficulty: Intermediate

Objective

Evaluate a video conferencing room setup for hybrid learning equity and accessibility. Assess camera, microphone, speaker, and platform features from the perspective of both in-room and remote participants.

Materials

- Video conferencing system (Zoom, Microsoft Teams, Google Meet, or equivalent)
- Camera (built-in or external USB webcam, wide-angle preferred)
- Speakerphone, USB speaker, or room speaker and microphone array

- Second device for remote participant view (phone, second laptop, or partner at remote location)
- Video Conferencing Room Audit Form (this worksheet)

Privacy and Recording Notice

Obtain instructor permission and all participant consent before recording any portion of this lab session. If using a second participant to simulate a remote attendee, that person must be informed they are participating in a lab evaluation. Do not share recordings externally. Delete test recordings after the lab is complete per your institution's data retention policy.

Pre-Lab Preparation

- Review Chapter 7 vocabulary: *codec, field of view (FOV), auto-framing, echo cancellation, noise suppression, packet loss, jitter, captioning, hybrid equity, speaker identification*.
- Arrange for a second device or partner before lab begins. If unavailable, use a recording-based alternative (record yourself, play back, and evaluate).

Procedure

Step 1 — Initiate Test Meeting: Launch a video conferencing session with a remote participant (or recording mode). Confirm both camera and microphone are active. Allow 3 minutes for the system to stabilize before beginning evaluation.

Step 2 — Evaluate Camera Field of View: Walk to the far left and far right of the room. Confirm whether the remote participant can see you throughout. Estimate the camera's horizontal field of view in degrees (wide-angle = $\sim 110^\circ$, standard = $\sim 78^\circ$, narrow = $\sim 60^\circ$).

Step 3 — Test Auto-Framing: If the camera system includes auto-framing (e.g., AI tracking), walk around the room while speaking. Observe how quickly and accurately the camera tracks. Rate as Excellent / Good / Fair / Poor / Not Available.

Step 4 — Evaluate Microphone Coverage: Have the remote participant close their eyes (or review recording). Walk to the far end of the table or room and speak in a normal voice. Ask the remote participant to rate intelligibility from the far position (1-5). Repeat from near position.

Step 5 — Measure Echo and Reverb: Clap hands sharply once at the far end of the room. Listen for audible echoes or reverb from the speaker. Ask the remote participant to describe what they heard. Rate echo/reverb as Severe / Noticeable / Minimal / None.

Step 6 — Evaluate Content Sharing Quality: Share your screen showing a presentation slide with 24pt text. Ask the remote participant to rate legibility of shared content (1-5).

Step 7 — Test Captioning: Enable the platform's automated captioning feature. Speak three complex sentences from the textbook. Review captions for accuracy. Rate caption accuracy as Excellent / Good / Fair / Poor.

Step 8 — Evaluate Speaker Identification: If the platform supports speaker labels (name displayed when speaking), test whether the correct participant name is attributed to each speaker.

Step 9 — WCAG 2.1 Level AA Compliance Review: Review the platform's captioning, screen reader compatibility, and keyboard navigation. Document items that comply and items that do not meet WCAG 2.1 Level AA standards.

Step 10 — Rate Remote Participant Equity: From the perspective of the remote participant, rate the overall equity of their experience compared to in-room participants. Use the rating scale and Hybrid Equity Gap Analysis table below.

Worksheet: Video Conferencing Room Audit Form



Room Audit Rating (1 = Very Poor, 5 = Excellent)

Criterion	Rating (1-5)	Evidence / Observation	Accessibility Implications
Camera Field of View (coverage of			

Criterion	Rating (1-5)	Evidence / Observation	Accessibility Implications
entire room)			
Framing Quality (instructor/presenter centered)			
Auto-Framing Performance			
Microphone Coverage (near position)			
Microphone Coverage (far position)			
Echo / Reverb Control			
Content Share Legibility			
Automated Caption Accuracy			
Speaker Identification / Attribution			
Remote Participant Equity (overall)			
Ease of Use (join, share, camera controls)			

Hybrid Equity Gap Analysis

Issue Found	Impact (High/Med/Low)	Who Is Affected	Recommended Fix	Est. Cost (Low/Med/High)

Issue Found	Impact (High/Med/Low)	Who Is Affected	Recommended Fix	Est. Cost (Low/Med/High)

Reflection Questions



Rubric Reference:

See Lab 7 Rubric in the Instructor Guide. Key dimensions: Audit form completeness (30%), Gap analysis depth (30%), Reflection quality and ISTE connection (40%).

LAB 8: Classroom AV Design Sketch

Corresponding to Chapter 8 • Estimated Time: 90 Minutes • Difficulty: Intermediate–Advanced

Objective

Create a basic AV system design for a specified classroom scenario. Apply sight line calculations, display sizing rules, speaker placement principles, camera placement for hybrid, and cable routing to produce a scaled floor plan with annotated bill of materials.

Materials

- Graph paper (provided) or free design tool: [Floorplanner.com](https://www.floorplanner.com)
- Pencil, ruler, and eraser
- Calculator
- AV component reference (Appendix B)

Lab Scenario (Required)

Room: Rectangular classroom, 30 ft wide × 40 ft deep (9.1 m × 12.2 m)

Occupancy: 25 students + 1 instructor

AV Requirements: Primary display for instruction, audio reinforcement for

hybrid and accessibility, camera for hybrid learning, AV control system, accessible input connectivity for educator

Constraints: Windows on the right side wall (glare considerations). Existing whiteboard on front wall (center). Budget: moderate (no custom infrastructure).

Pre-Lab Preparation

- Review Chapter 8 vocabulary: *throw ratio*, *viewing distance*, *sight line*, *4/6/8 rule*, *coverage angle*, *speaker delay*, *cable routing*, *bill of materials (BOM)*, *ADA reach range*.
- Memorize the key design rules: **1/6 rule** (front row maximum: viewing distance = screen height $\times$ 6); **1/8 rule** (back row maximum: screen height $\times$ 8); **4/6/8 flat panel rule** (diagonal inches = room depth in feet $\div$ constant).

Procedure

Step 1: Using the graph paper grid below (scale: 1 square = 2 ft), draw the room to scale. Mark windows, doors, and the existing whiteboard.

Step 2: Mark window locations and indicate the direction of potential glare. Note: place primary display on the wall opposite windows if possible to minimize ambient light conflict.

Step 3: Determine display placement. Decide between projector or flat panel as primary display. Apply the 4/6/8 rule for flat panel sizing OR calculate throw ratio for a projector. Record calculations in the Sight Line Calculation template.

Step 4: Draw the display on the floor plan at the chosen location. Draw sight lines from the first row and last row of seating. Verify all students have unobstructed views.

Step 5: Place speakers on the floor plan. For a 40-ft deep room, two front-fill ceiling or wall speakers at approximately 10 ft depth, plus a center cluster or distributed

Category	Qty	Specification Description	Est. Unit Cost Range	Purpose
Primary Display				
Display Mount				
Speakers (front)				
Speaker Amplifier				
Microphone (educator)				
Camera (hybrid)				
AV Switcher / Scaler				
Control System / Interface				
Input Connectivity Panel				
Cable Package (HDMI, Cat6, Speaker, etc.)				
Power (PDU / Surge Protection)				
Installation / Labor (est.)	1 lot			

Reflection Questions



Rubric Reference:

See Lab 8 Rubric in the Instructor Guide. Key dimensions: Scaled floor plan accuracy (30%), Calculations (25%), Bill of materials (20%), Reflection quality (25%).

LAB 9: Digital Signage Content Planning

Corresponding to Chapter 9 • Estimated Time: 90 Minutes • Difficulty: Intermediate

Objective

Plan and mock up an accessible, well-structured digital signage content strategy for an educational setting. Apply WCAG accessibility guidelines to content design and produce two sample slides with an accompanying content calendar and governance policy outline.

Materials

- Computer with Canva (canva.com — free), Google Slides, or PowerPoint
- WebAIM Contrast Checker (webaim.org) — browser-based, free
- Content Calendar Template (this worksheet)
- Printer or screen for peer presentation

Pre-Lab Preparation

- Review Chapter 9 vocabulary: *digital signage, content management system (CMS), content zone, dwell time, refresh rate, WCAG, color contrast ratio, emergency alert integration, wayfinding, content governance*.
- Identify the audience and location for your signage before lab begins (e.g., main hallway at a high school; library entrance at a community college).

Write it below.

Pre-Lab: Audience and Location Definition

Audience: _____ Location: _____

Approximate daily viewership: _____ people Average viewing distance:
_____ feet

Ambient light conditions: _____ Display orientation:
Landscape / Portrait

Procedure

Step 1: Define your signage audience and location (complete Pre-Lab above).
Identify at least 3 distinct audience segments who will view this signage.

Step 2: Research WCAG accessibility requirements for digital signage. Key items: minimum color contrast ratio, minimum font size for viewing distance, alternative text for non-text content, and emergency alert requirements. Record key findings in the Content Accessibility Checklist.

Step 3: Define your 4 content zones and their purpose in the Content Plan table:
(a) Emergency/Safety zone, (b) Scheduled Events zone, (c) Wayfinding zone, (d) Student/Community Content zone.

Step 4: Create a content calendar for one week (Monday–Friday). Assign content to each zone for each day. Specify dwell time per slide (typically 8–15 seconds for general content; emergency content should be persistent).

Step 5: Design Sample Slide A: a Scheduled Events announcement slide. Use your chosen tool (Canva, Slides, PowerPoint). Apply the following: minimum 30pt font for 10-foot viewing distance, accessible color palette, no information conveyed by color alone, all graphics have alt-text descriptions (note in your plan).

Step 6: Design Sample Slide B: a Wayfinding slide with directional information for your location. Apply the same accessibility requirements.

Step 7: For each sample slide, use WebAIM Contrast Checker to verify text vs. background contrast ratios. Record results in the Accessibility Checklist.

Step 8: Check text size for viewing distance: at 10 feet, minimum recommended font height is approximately 1 inch ($\approx 72\text{pt}$ for most screens). Verify your slides meet or exceed this at the intended display size.

Step 9: Create a brief Content Submission Policy outline: who can submit content, how far in advance, approval workflow, prohibited content categories, and emergency override procedure.

Step 10: Present your content plan and two sample slides to the class. Explain your accessibility decisions and content calendar structure. Receive peer feedback using the Peer Evaluation Form.

Worksheet: Digital Signage Content Plan

Content Zone Definitions

Zone	Purpose	Priority (1=Highest)	Default Dwell Time	Who Manages Content	Override Behavior
A — Emergency/Safety	Life-safety alerts, emergency information	1	Persistent until cleared		Override all other zones immediately
B — Scheduled Events		2			
C — Wayfinding		3			
D — Student/Community Content		4			

Weekly Content Calendar

Day	Zone A (Emergency)	Zone B (Events)	Zone C (Wayfinding)	Zone D (Community)	Rotation Time
Monday	No active alerts				
Tuesday	No active alerts				
Wednesday	No active				

Day	Zone A (Emergency)	Zone B (Events)	Zone C (Wayfinding)	Zone D (Community)	Rotation Time
	alerts				
Thursday	No active alerts				
Friday	No active alerts				

Content Accessibility Checklist

Criterion	Requirement	Slide A Result	Slide B Result	Pass/Fail
Text- Background Contrast Ratio	$\geq 4.5:1$ (normal text) / $\geq 3:1$ (large text)			
Minimum Font Size (10 ft viewing)	≥ 30 pt for slide, ≥ 72 pt for 10-ft screen			
Color not sole conveyor of information	Icons, labels, or patterns used alongside color			
All images have alt-text descriptions noted	Alt text documented in design notes			
No flashing/strobing content	No content flashing > 3 times per second			
Emergency content overrides all zones	Policy states override is automatic/im mediate	N/A — Policy item	N/A	

Content Submission Policy Outline

Policy Item	Your Policy Statement
Who may submit content	
Advance submission deadline	
Approval authority	
Prohibited content categories	
Emergency override procedure	
Content removal process	

Reflection Questions



Rubric Reference:

See Lab 9 Rubric in the Instructor Guide. Key dimensions: Content plan completeness (25%), Accessibility checklist (25%), Sample slides quality (25%), Reflection and peer evaluation (25%).

LAB 10: AV Troubleshooting Challenge

Corresponding to Chapter 10 • Estimated Time: 90 Minutes • Difficulty: Intermediate-Advanced

Objective

Apply a systematic troubleshooting methodology to diagnose and resolve a minimum of three instructor-introduced AV system faults. Document the troubleshooting process using a structured log and write a fault-prevention procedure.

Materials

- AV system with 3+ instructor-introduced faults (see Instructor Setup Note below)
- Troubleshooting Log Form (this worksheet)
- Multimeter (optional)
- Timer or clock

- Reference: Signal flow diagram from Lab 4 (if available)

Instructor Setup Note

Before the lab session, introduce at least 3 of the following faults into the AV system. Do not inform students which faults are present. Suggested faults: (1) Wrong input source selected on the AV switcher; (2) Muted audio channel on the mixer; (3) Display output resolution set to incorrect/unsupported value; (4) HDMI cable disconnected at the back of the display; (5) IP address conflict (two devices assigned same static IP); (6) Projector lamp mode set to economy/off; (7) Control system programmed to wrong source label. Document which faults were introduced on a separate instructor answer key — do not attach to this workbook.

Safety During Troubleshooting

Always power down the system before replacing cables or accessing rear panel connections. Do not use a multimeter on AC power circuits. If you suspect an electrical fault (burning smell, popping sound, tripped breaker), do NOT attempt to diagnose — power down immediately and notify the instructor.

Pre-Lab Preparation

- Review Chapter 10 vocabulary: *systematic troubleshooting, OSI model, physical layer, symptom, hypothesis, root cause, isolation, signal injection, time-to-resolution, fault report.*
- Review the AV troubleshooting model: Physical Layer → Signal Layer → Configuration/Software Layer → User/Training Layer.

Procedure

Step 1 — Document System State: Before any troubleshooting, document the observable symptoms (what does the system do / not do). Do not touch any controls yet — just observe and record in the Troubleshooting Log.

Step 2 — Apply the Physical Layer First: Check all physical connections. Are all cables fully seated? Are all devices powered on? Is any indicator light showing an error state? Check visually and document.

Step 3 — Check Signal Path: Follow the signal flow from source to output. Use your Lab 4 signal flow diagram if available. At each junction, determine if the signal is present (by checking indicator LEDs, meter, or output device response).

Step 4 — Check Software and Settings: Check source selection (correct input?), volume/mute status, output resolution settings, IP configuration, and control system programming.

Step 5 — Form a Hypothesis: Based on Steps 2–4 observations, state your hypothesis for each fault: "I believe the problem is [X] because [evidence Y]." Record each hypothesis in the log before testing.

Step 6 — Test the Hypothesis: Make one change at a time. After each change, observe the result. If the problem is not resolved, document and move to the next hypothesis. Do NOT make multiple changes simultaneously.

Step 7 — Document Resolution: When a fault is resolved, record exactly what the root cause was and what corrective action resolved it.

Step 8 — Calculate Time-to-Resolution: Record the start time for each fault's troubleshooting and the time of resolution. Calculate total elapsed time per fault.

Step 9 — Complete the Post-Fault Report: For each resolved fault, complete the fault report template below.

Step 10 — Write a Prevention Procedure: For each fault type identified, write a brief preventive procedure (e.g., checklist item, policy, or maintenance schedule) that would reduce the likelihood of recurrence.

Worksheet: Troubleshooting Log



Troubleshooting Log — Complete one row per test/hypothesis

Fault #	Symptom Observed	Layer (Physical / Signal / Config)	Hypothesis	Test Performed	Result	Resolved? (Y/N)	Time Elapsed (min)
1							
1							
2							
2							
3							
3							
4							

 Fault Report (complete one per resolved fault)

Field	Fault 1	Fault 2	Fault 3
Symptom (what was wrong?)			
Root Cause			
Troubleshooting Layer (Physical / Signal / Config)			
Corrective Action			
Time to Resolution (minutes)			
Prevention Procedure			

 Troubleshooting Summary

Metric	Value
Total faults introduced by instructor	
Total faults resolved	

Metric	Value
Total faults not resolved (explain in reflection)	
Average time-to-resolution per fault (min)	
Most common troubleshooting layer (Physical / Signal / Config)	

Reflection Questions

Rubric Reference:

See Lab 10 Rubric in the Instructor Guide. Key dimensions: Troubleshooting log rigor (35%), Fault reports (25%), Prevention procedures (15%), Reflection quality (25%).

LAB 11: Accessibility Audit

Corresponding to Chapter 11 • Estimated Time: 100 Minutes • Difficulty: Advanced

Objective

Conduct a comprehensive AV accessibility audit of a real or simulated learning space. Apply ADA Standards, Section 508, and WCAG 2.1 requirements. Produce a prioritized remediation plan and an administrative summary memo.

Materials

- AV Accessibility Audit Checklist (this worksheet — 50 items)
- Physical room to audit (or virtual campus tour as alternative)
- ADA Standards for Accessible Design reference (publicly available from [ADA.gov](https://www.ada.gov))
- WCAG 2.1 Quick Reference ([w3.org/WAI/WCAG21/quickref/](https://www.w3.org/WAI/WCAG21/quickref/))
- Hearing loop test app (free: Loop Check Pro or equivalent) if assistive listening system is present

- Measuring tape
- Camera for documentation photos

Role-Play Guidance for Step 8

Step 8 involves role-playing as (or interviewing) a user with a disability. Conduct this exercise with sensitivity and respect. The goal is to understand, not to stereotype. If a student with a disability is willing to share their own experience, that input is valuable and should be documented with their permission. Role-play must never involve mockery or caricature.

Pre-Lab Preparation

- Review Chapter 11 vocabulary: *ADA, Section 508, WCAG, assistive listening system (ALS), hearing loop (induction loop), telecoil, captioning, audio description, reach range, accessible route, remediation.*
- Review ADA Standards Chapter 7 (Communication Elements and Features) and Chapter 2 (Scoping Requirements) before the lab.

Procedure

Step 1: Review ADA and WCAG 2.1 requirements relevant to AV systems. Focus on: assistive listening, captioning, reach ranges for controls, accessible routes, and display height/placement.

Step 2: Conduct a physical inspection of the room using the 50-item Audit Checklist below. Mark each item as Compliant, Non-Compliant, or N/A. Add notes for all non-compliant items.

Step 3: Test the assistive listening system (hearing loop, FM, or IR) if present. Use a hearing loop test app or a T-coil enabled hearing aid. Rate coverage and signal quality.

Step 4: Test captioning on all displays in the room. Enable captions on each source type (computer content, live camera, video playback). Rate accuracy and availability.

Step 5: Evaluate hearing loop coverage with a test device or app (if loop is present). Walk the perimeter and seating area. Note any dead zones.

Step 6: Check AV-related signage for accessibility: tactile signage at room entrance, high-contrast wayfinding, hearing loop indicator sign (if applicable).

Step 7: Evaluate the control system interface for accessibility (refer to Lab 6 findings if this is the same room).

Step 8: Conduct a role-play or interview: simulate being a first-time user of the room who is deaf, hard-of-hearing, or low-vision. Attempt to: enter the room, understand what displays are showing, hear the audio content, and control the AV system independently. Document experiences and barriers encountered.

Step 9: Document all non-compliant or gap items. For each, assign a priority (High = impacts safety or core accessibility, Medium = significant impact, Low = enhancement).

Step 10: Select the top 5 remediation items. For each, estimate a cost range (Low: under \$500, Medium: \$500–\$5,000, High: over \$5,000).

Step 11: Complete the Remediation Priority Matrix.

Step 12: Write a one-page summary memo to the building administrator using the Administrative Memo Template below.

Worksheet: AV Accessibility Audit Checklist (50 Items)

 **Rating: C = Compliant | NC = Non-Compliant | N/A = Not Applicable**

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
SECTION 1: Physical Accessibility of AV Controls					

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
1	AV control panel height is 15-48 inches from floor (forward reach) or 15-48 inches (side reach)	ADA 308			
2	Operable parts require no tight grasping, pinching, or twisting of wrist	ADA 309.4			
3	30×48 inch clear floor space provided at AV control location	ADA 305			
4	AV rack/educator station is on an accessible route (36 in min. aisle)	ADA 402			
5	Input connectivity panels are within ADA reach range at educator station	ADA 308			
6	Touch screen controls (if used) have tactile or	ADA 707.6			

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
	alternative non-visual access mode				
7	AV equipment controls are operable with one hand	ADA 309.4			
8	Power switches and volume controls are clearly identifiable (tactile label or color)	508 §1194.31			
SECTION 2: Assistive Listening Systems					
9	Assistive listening system (ALS) is present (hearing loop, FM, or IR)	ADA 219			
10	ALS type signage is posted at room entrance with ISA symbol and ear symbol	ADA 216.10			
11	Hearing loop coverage extends throughout the primary seating area	IEC 60118-4			

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
12	Hearing loop signal level is within ± 3 dB of target throughout coverage area	IEC 60118-4			
13	FM or IR receiver units are available in sufficient quantity (minimum: 4% of fixed seating)	ADA 219.3			
14	Receivers are maintained, charged, and operational	Best Practice			
15	ALS is compatible with hearing aids using T-coil (telecoil)	ADA 706.2			
SECTION 3: Captioning and Audio Description					
16	Closed captioning is available and functional on all displays for live content	WCAG 1.2.4			
17	Closed captioning is available for recorded	WCAG 1.2.2			

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
	video content				
18	Captions are legible from the farthest seat in the room (font size, contrast)	Best Practice			
19	Caption accuracy is at or above 98% for pre-recorded content	ADA Title II			
20	Captions do not obscure essential on-screen content	WCAG 1.4.2			
21	Audio description track is available for visual-only content (video)	WCAG 1.2.5			
22	Video conferencing platform used in this room supports live captioning	WCAG 1.2.4			
23	Digital signage in or adjacent to room includes text	WCAG 1.1.1			

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
	versions of all visual content				
SECTION 4: Display Accessibility					
24	Primary display visible from all required seating positions without obstruction	Best Practice			
25	Display text contrast meets WCAG AA minimum (4.5:1 for body text)	WCAG 1.4.3			
26	Minimum font size on presenter materials legible from back row	Best Practice			
27	Display is not positioned in front of a bright light source causing glare	Best Practice			
28	Interactive display (if present) is reachable from wheelchair position	ADA 308			
29	Display does not	WCAG 2.3.1			

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
	flicker (rate < 3 Hz) which can trigger photosensitive conditions				
30	Secondary display or confidence monitor available for speaker reference	Best Practice			
SECTION 5: Microphone and Audio Accessibility					
31	Wearable or handheld microphone available so presenter is not tethered to podium	Best Practice			
32	Audio system volume is independently adjustable (not locked at a single level)	Best Practice			
33	Audio system capable of achieving 65–70 dB SPL at rear seating with speech reinforcement	ANSI S12.60			

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
34	Signal-to-noise ratio of audio system is sufficient (background noise < NC35)	ANSI S12.60			
35	Audio system output feeds the assistive listening system (hearing loop or FM transmitter)	ADA 706.1			
36	Microphone system does not require fine motor skill to operate (push-to-talk, mute button accessible)	ADA 309.4			
SECTION 6: Video Conferencing Accessibility					
37	Video conferencing platform supports keyboard navigation (no mouse required)	WCAG 2.1.1			
38	Screen reader compatibility confirmed for platform in use	WCAG 4.1.3			

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
39	Camera position allows clear view of sign language interpreters (if applicable)	Best Practice			
40	Spotlight or pinning feature available to feature a sign language interpreter	Best Practice			
41	Platform allows captioning display to be repositioned on screen	WCAG 1.4.8			
42	Bandwidth sufficient to maintain 720p video and audio without interruption	Best Practice			
SECTION 7: Physical Room Accessibility					
43	Accessible seating (wheelchair space) is integrated with general seating, not segregated	ADA 221.3			
44	Accessible seating has companion seat	ADA 221.3			

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
	immediately adjacent				
45	Lighting in the room is sufficient for lip-reading and sign language interpretation	Best Practice			
46	Room acoustics (reverberation time) is appropriate for speech intelligibility	ANSI S12.60			
47	HVAC / mechanical noise does not interfere with speech intelligibility (NC-35 or better)	ANSI S12.60			
48	Emergency alert system (visual strobe + audio) is present and functional	ADA 702			
49	Emergency visual alerts (strobes) are visible from all seating	ADA 702.3.3			

#	Audit Item	ADA / WCAG Ref.	C / NC / N/A	Priority H/M/L	Notes
	positions				
50	Staff in this room are trained on how to use assistive listening system and activate captions	Best Practice			

 Remediation Priority Matrix (Top 5 Items)

Rank	Audit Item #	Description	ADA/ WCAG Ref.	Priority H/M/L	Est. Cost Range	Responsible Party
1						
2						
3						
4						
5						

Administrative Memo Template

 Memo to Administration — AV Accessibility Audit Findings

TO: Building Principal / Director of Facilities

FROM:

DATE:

SUBJECT: AV Accessibility Audit — [Room Name/Location]

Purpose of Audit:

Summary of Findings:

Items in Compliance: ____ of 50 items audited

Non-Compliant Items Identified: ____

Top 3 Recommended Actions (brief):

Estimated Total Remediation Cost Range: \$_____ - \$_____

Requested Next Step:

Reflection Questions



Rubric Reference:

See Lab 11 Rubric in the Instructor Guide. Key dimensions: Audit checklist thoroughness (30%), Remediation matrix (25%), Administrative memo quality (20%), Reflection depth and advocacy argument (25%).

LAB 12: Emerging Technology Exploration

Corresponding to Chapter 12 • Estimated Time: 90 Minutes + Presentation •
Difficulty: Advanced

Objective

Research an emerging AV technology relevant to educational settings. Evaluate its current maturity, educational applications, accessibility and equity implications, cost scalability, and alignment with ISTE Standards and AVIXA professional domains. Produce and deliver a 5-minute class presentation with written summary.

Materials

- Computer with internet access
- Presentation tool (PowerPoint, Google Slides, Canva, or equivalent)
- Emerging Technology Research Template (this worksheet)
- Peer Evaluation Form (this worksheet)



Choose One of the Following Emerging Technologies

- AI-Powered Classroom Systems (autonomous cameras, AI room management, occupancy sensing)
- Extended Reality (XR): AR, VR, or MR Headsets for Educational Use
- Digital Twins of Learning Spaces
- Spatial Audio and Object-Based Audio for Education
- AI-Powered Automatic Speech Recognition (ASR) for Live Captioning
- Autonomous Camera Framing Systems (e.g., AI-based presenter tracking)
- Other emerging technology — requires instructor pre-approval

Pre-Lab Preparation

- Review Chapter 12 vocabulary: *technology readiness level (TRL)*, *immersive technology*, *AI*, *XR*, *spatial audio*, *digital twin*, *ASR*, *equity gap*, *ISTE Standards*, *AVIXA CTS Domain*.

- Select your technology topic and notify your instructor before beginning research.
- Gather at least 3 credible sources (peer-reviewed journal, industry white paper, and news article).

Procedure

Step 1: Choose your emerging technology topic (from the list above or instructor-approved alternative).

Step 2: Research the current state of the technology. Define it precisely, describe how it works, and identify where it falls on the technology readiness level spectrum (TRL 1-9, or Early Adopter / Mainstream / Commoditized categories).

Step 3: Identify 3 or more specific educational use cases where this technology is already being used or piloted. Find real examples (schools, universities, or districts) if possible.

Step 4: Identify and analyze accessibility implications: Does this technology improve accessibility for students with disabilities? Does it create new barriers? Provide specific examples.

Step 5: Identify and analyze equity implications: Is this technology equally accessible to students in under-resourced schools? Does its cost, infrastructure requirement, or digital literacy demand create equity gaps?

Step 6: Analyze cost and scalability for both K-12 and higher education: approximate per-classroom cost, infrastructure requirements (network, power, space), ongoing maintenance costs, and total cost of ownership over 5 years.

Step 7: Identify the ISTE Standard(s) this technology most directly advances. Cite the specific standard number and indicator (e.g., ISTE Educator Standard 5: Designer, 5b).

Step 8: Identify the AVIXA CTS Domain (from: Needs Analysis, System Design, Project Management, Installation, Operation, Service, Customer Relationship) that this technology most significantly affects.

Step 9: Create a 5-minute presentation (5–8 slides) covering: technology overview, educational use cases, accessibility and equity analysis, cost/scalability, ISTE/AVIXA alignment, and your personal recommendation.

Step 10: Present to the class. Peers complete the Peer Evaluation Form during the presentation. Submit both the presentation file and a written summary (1–2 pages) to the instructor after presenting.

Worksheet: Emerging Technology Research Template

Research Documentation

Field	Your Response
Technology Name	
Precise Definition (2–3 sentences)	
How It Works (brief technical description)	
Current Maturity Level (Early Adopter / Mainstream / Commoditized)	
Technology Readiness Level (TRL 1–9)	
Educational Use Case 1	
Educational Use Case 2	
Educational Use Case 3	
Real-World Example (school/institution using this technology)	
Accessibility Features (how does it help students with disabilities?)	
Accessibility Gaps (what barriers does it create or fail to address?)	
Equity Implication 1 (cost, infrastructure, etc.)	
Equity Implication 2	
Approximate Per-Classroom Cost (USD)	

Field	Your Response
Infrastructure Requirements	
5-Year Total Cost of Ownership Estimate	
Scalability for K-12 (Low / Medium / High)	
Scalability for Higher Education (Low / Medium / High)	
ISTE Standard Advanced (number + indicator)	
AVIXA Domain Most Affected	
Source 1 (citation)	
Source 2 (citation)	
Source 3 (citation)	
Personal Recommendation (adopt now / pilot / monitor / avoid — and why)	

Peer Evaluation Form

 **Peer Evaluation — Complete during or immediately after each classmate's presentation**

Presenter Name: _____ **Technology Topic:**

Evaluator Name: _____

Criterion	Rating (1-5)	Specific Feedback (one sentence minimum)
Clarity of technology explanation		
Depth of educational use case analysis		
Quality of accessibility and equity analysis		

Criterion	Rating (1-5)	Specific Feedback (one sentence minimum)
Realism and depth of cost/scalability analysis		
ISTE and AVIXA connection accuracy		
Presentation clarity and organization		
Personal recommendation strength and justification		
Overall Rating		

One thing I learned from this presentation:

One question I would ask this presenter:

Reflection Questions



Rubric Reference:

See Lab 12 Rubric in the Instructor Guide. Key dimensions: Research template completeness (25%), Presentation quality (25%), Peer evaluations (15%), Written summary (15%), Reflection depth (20%).

APPENDIX A

Measurement Templates

Reproducible blank measurement logs for use across all labs. Make additional copies as needed.

A-1: SPL (Sound Pressure Level) Measurement Log

Lab #	Date	Room	Position Description	Distance from Source (ft/m)	Measured dB SPL	Ambient Noise Floor dB	Measurement App Used	Notes

A-2: Lux (Ambient Light) Measurement Log

Lab #	Date	Room	Measurement Location	Lights ON lux	Lights OFF lux	Blinds Open/Closed	Time of Day	Notes

A-3: Cable Continuity Log

Cabl e ID/L abel	Cabl e Type	Leng th (ft)	Con nect or A Type	Con nect or B Type	Cont inuit y Test Resu lt (Pas s/Fai l)	Pin 1	Pin 2	Pin 3	Grou nd	Note s

A-4: IP Address Assignment Log

Devic e Nam e	Devic e Type	MAC Addr ess	IP Addr ess (Stati c or DHCP)	Subn et Mask	Gate way	VLAN	Port (Swit ch)	Date Assig ned	Note s

AV Equipment Quick-Reference Card

Common connectors, cable types, and signal types for educator reference.

Reproduce and keep at your workstation.

Common AV Connectors

Connector Name	Signal Type	Pins / Format	Common Uses in Education	Lock Mechanism
HDMI Type A	Digital Video + Audio	19-pin	Laptop to display, Blu-ray to projector	Friction (no lock)
HDMI Type D (Micro)	Digital Video + Audio	19-pin micro	Camera to display	Friction
DisplayPort	Digital Video + Audio	20-pin	PC to monitor, daisy chaining	Latching clip
USB-C (with DP Alt)	Digital Video + Audio + Data + Power	24-pin	Modern laptops to displays	Friction
VGA (HD-15)	Analog Video only	15-pin D-Sub	Legacy projectors and monitors	Thumbscrews
XLR (3-pin F/M)	Balanced Analog Audio	3-pin	Microphones, mixers, amplifiers	Twist-lock
TRS (1/4")	Balanced Analog Audio (stereo or balanced mono)	Tip-Ring-Sleeve	Line-level audio, headphones, instruments	Friction
TS (1/4")	Unbalanced Analog Audio (mono)	Tip-Sleeve	Unbalanced line connections, instruments	Friction

Connector Name	Signal Type	Pins / Format	Common Uses in Education	Lock Mechanism
RCA (Phono)	Unbalanced Analog Audio or Composite Video	Single pin	Consumer audio, legacy video	Friction
3.5mm TRS (1/8")	Stereo Analog Audio	Tip-Ring-Sleeve	Laptop headphone/mic jack, consumer devices	Friction
RJ-45 (Cat5e/Cat6)	Network Data (Ethernet)	8-pin	Network AV, HDBaseT, PoE devices	Snap latch
BNC	Analog/Digital Video or RF	Coaxial	Legacy video, broadcast, SDI signals	Twist-lock bayonet
SDI (BNC)	Serial Digital Interface (HD video)	Coaxial BNC	Professional video, broadcast	Twist-lock bayonet
Euroblock (Phoenix)	Balanced/Unbalanced Audio	Screw terminal block	Fixed AV installations, DSP connections	Screw terminal
USB-A / USB-B	Data + Power	4 or 9-pin	USB audio interfaces, webcams, storage	Friction

Common AV Cable Types and Characteristics

Cable Type	Signal Carried	Typical Max Run (passive)	Shielded?	Notes for Educators
HDMI Standard	Digital video + audio	15 ft (5 m) passive; 50 ft+ with active	Yes	Most common classroom cable — check version (2.0 for 4K)
Cat6 Ethernet	Network data;	328 ft (100 m)	Optional (STP)	Used for both

Cable Type	Signal Carried	Typical Max Run (passive)	Shielded?	Notes for Educators
	HDBaseT	for 1G Ethernet		networking and AV extension (HDBaseT)
XLR Balanced	Balanced analog audio	200+ ft without signal loss	Yes	Professional standard for microphones; balanced = noise rejection
Speaker Wire	Amplified audio (power)	Depends on gauge and impedance	No	Connect amplifier to passive speakers only — not mic-level
Optical (TOSLINK)	Digital audio	~33 ft (10 m)	Optical — immune to EMI	Used for surround sound from consumer devices
Coax (RG-6)	RF, SDI video, composite	Hundreds of feet for RF	Yes	Cable TV, antenna, legacy surveillance, broadcast SDI
USB 3.x	Data + power	~15 ft (5 m) passive	Partial	Active USB or USB-over-IP for longer runs
DisplayPort 1.4	Digital video + audio	~6 ft (2 m) at 4K 60Hz; 15 ft at 1080p	Yes	Supports daisy chaining multiple monitors
Fiber Optic (HDMI/AV)	Digital video/audio over fiber	1,000+ ft	Optical — immune to EMI	Long-distance AV runs; unidirectional

Signal Types Quick Reference

Signal Type	Analog or Digital	Level	Typical Connectors	Notes
Microphone Level	Analog	-40 to -60 dBV	XLR, 3.5mm TRS	Very low level — needs preamplification
Line Level (consumer)	Analog	-10 dBV	RCA, 3.5mm TRS	Standard for consumer AV equipment
Line Level (professional)	Analog	+4 dBu	XLR, 1/4" TRS	Standard for professional AV and broadcast
Speaker Level	Analog (amplified)	Variable (watts)	Speaker wire terminals	Never connect to mic or line inputs — will damage equipment
HDMI	Digital	N/A (digital)	HDMI	Carries video, audio, and HDCP content protection
HDBaseT	Digital (over Cat cable)	N/A	RJ-45	HDMI + power + control over a single Cat6 cable up to 330 ft
SDI	Digital	N/A	BNC	Professional broadcast digital video over coax
IP (Network)	Digital (packetized)	N/A	RJ-45, SFP	AV over IP (AVoIP) — video, audio, and control over Ethernet
Control (RS-232)	Serial Digital	N/A	DB-9, 3.5mm, screw terminal	Device control protocol — used by control systems to command AV devices

APPENDIX C

Lab Safety Contract

AV Fundamentals for Educators — Lab Safety Contract

This contract must be signed by the student and retained by the instructor before the student participates in any lab session involving powered AV equipment.

I, the undersigned student, acknowledge and agree to the following:

1. I have read and understand the General AV Lab Safety Rules in this workbook (pages 4–5).
2. I will complete the Universal Safety Checklist at the beginning of every lab session involving powered equipment.
3. I will not power on any AV equipment without the instructor or authorized lab monitor present.
4. I will not bring liquids of any kind near powered AV equipment.
5. I will immediately report any damaged cables, equipment malfunctions, unusual smells, or safety hazards to the instructor without attempting to fix them myself.
6. I understand that sound pressure levels will be kept below 85 dB SPL at all times, and I will use hearing protection when it is made available.
7. I will never scan any network without explicit instructor authorization and will limit network activities to the designated lab subnet.
8. I understand that any student behavior that creates an unsafe condition

may result in removal from the lab for that session and may affect my grade.

9. I will treat all AV equipment with care, report any damage immediately, and not remove equipment from the lab without instructor permission.

10. I understand that the safety rules apply to me at all times during lab sessions, even if I have prior experience with AV equipment.

Student Name (Print):

Student Signature: **Date:**

Course Name: AV Fundamentals for Educators **Section:**

Institution:

Instructor Acknowledgment:

I confirm that the student above has received a safety orientation covering the rules in this workbook and has demonstrated understanding of the emergency stop procedure, hearing protection requirements, and network usage limitations.

Instructor Name (Print):

Instructor Signature: **Date:**

APPENDIX D

Glossary of Lab Terms

50 terms specific to hands-on AV lab practice. Definitions are written for educators, not technicians.

1. Ambient light

Light already present in a room from windows, ceiling lights, or other sources that is not part of the AV system. High ambient light reduces the apparent brightness of projected images.

2. Amplifier (power amplifier)

A device that increases the level of an audio signal from line level to speaker level, providing power to drive speakers. In classroom AV, amplifiers may be integrated into the speaker or control system.

3. Audio interface

A device that converts analog audio signals (from microphones, instruments) into digital signals that a computer can process. Often connects via USB and replaces a traditional mixer for simple setups.

4. Balanced signal

An audio signal transmitted on two conductors plus a ground. Because noise picked up on the cable is cancelled when the signals are combined at the receiving device, balanced connections are preferred for long cable runs. XLR and TRS cables carry balanced signals.

5. Bandwidth

In networking, the maximum data transfer rate of a network connection, measured in megabits per second (Mb/s) or gigabits per second (Gb/s). In AV, higher bandwidth supports higher-resolution video and better audio quality.

6. Bill of materials (BOM)

A structured list of all components, cables, and accessories required to build or install an AV system, including quantities and descriptions. In education, a BOM is a key document for budget requests.

7. Cardioid

A microphone polar pattern that captures sound primarily from the front and sides while rejecting sound from the rear. Named for its heart-shaped sensitivity pattern. The most common pattern for educational use.

8. Codec

Short for "coder-decoder." A device or software that compresses and decompresses digital video and audio for transmission or storage. Video conferencing systems use codecs to transmit live video efficiently over a network.

9. Color contrast ratio

A mathematical measurement of the difference in luminance between foreground and background colors. WCAG 2.1 requires a minimum contrast ratio of 4.5:1 for normal text to be accessible to people with low vision.

10. Condenser microphone

A microphone type that uses a capacitor (condenser) element and requires phantom power (+48V) to operate. Generally more sensitive and detailed than dynamic microphones; commonly used in studio recording and conference room applications.

11. Control system

A centralized processor that sends commands to AV devices to automate room functions such as powering on displays, selecting inputs, and adjusting volume. Interacted with via a touchpanel, button panel, or app.

12. dBFS (decibels Full Scale)

A digital audio measurement where 0 dBFS represents the maximum possible digital signal level. All digital recordings should stay below 0 dBFS (negative values) to prevent clipping/distortion.

13. dBVU (Volume Units)

A traditional analog audio level measurement scale used on VU meters. A target level of -12 to -6 VU is typical for speech recording, leaving headroom for peaks.

14. DHCP (Dynamic Host Configuration Protocol)

A network service that automatically assigns IP addresses to devices when they connect to a network. AV devices may use DHCP or be assigned fixed (static) IP addresses for reliability.

15. Digital signage

Electronic display systems used to show dynamic content such as announcements, wayfinding, and emergency alerts in hallways, lobbies, and classrooms.

16. DSP (Digital Signal Processor)

An audio processor that applies digital effects — equalization, delay, compression, feedback suppression — to audio signals. In education, DSPs manage audio routing and processing for lecture halls and conferencing rooms.

17. Dwell time

In digital signage, the amount of time a content slide is displayed before transitioning to the next slide. Typically 8–15 seconds for general content.

18. Dynamic microphone

A microphone that works by electromagnetic induction — sound moves a diaphragm

attached to a coil of wire suspended in a magnetic field. Durable and feedback-resistant; the standard choice for handheld educator and presenter microphones.

19. Echo cancellation

A digital processing technique that removes the sound of a room's own speakers from the microphone signal during a video call, preventing remote participants from hearing a delayed copy of their own voice.

20. EQ (Equalization)

The process of adjusting the relative levels of specific frequency ranges in an audio signal. Boosting high frequencies can improve speech intelligibility; cutting low frequencies can reduce rumble and HVAC noise.

21. Feedback (audio)

The unwanted loop of sound that occurs when a microphone picks up its own amplified output from a speaker. Produces a ringing or howling tone. Managed through microphone placement, gain control, and equalization.

22. Field of view (FOV)

The angular width of the scene captured by a camera lens, measured in degrees. Wide-angle cameras (90°–120° FOV) capture more of a room; narrow cameras (40°–60° FOV) focus on a specific area.

23. Firmware

The permanent software embedded in an AV device (projector, switcher, camera, control system) that controls its basic functions. Firmware updates add features and fix security vulnerabilities.

24. Gain

An increase in signal level, measured in decibels (dB). Setting input gain correctly on a mixer or audio interface is the first step in creating a clean, noise-free audio signal chain.

25. HDBaseT

A technology standard that transmits uncompressed HDMI video, audio, power (PoE), and control signals over a single Cat5e/Cat6 Ethernet cable for up to 100 meters (328 ft). Widely used in classroom AV installations.

26. Hearing loop (induction loop)

An assistive listening system that transmits audio signals as a magnetic field around a defined loop of wire installed in a room. Users with hearing aids or cochlear

implants set to telecoil (T-coil) mode receive the signal directly. Required by ADA in assembly areas of a certain size.

27. Heuristic evaluation

A usability inspection method in which reviewers assess an interface against a set of recognized design principles (heuristics) to identify usability problems without requiring user testing. In this course, applied to AV control system touchpanels.

28. Hybrid equity

The principle that remote learners should have an equivalent — not merely adequate — educational experience compared to in-person participants. Requires attention to camera framing, microphone coverage, captioning, and content sharing quality.

29. IP address

A unique numerical label assigned to each device on a network. IPv4 addresses appear as four groups of numbers separated by periods (e.g., 192.168.1.50). AV devices on a school network each require a unique IP address to communicate.

30. Latency

The delay between a signal being sent and received. In video conferencing, latency above approximately 150 milliseconds becomes noticeable as conversation awkwardness. In AV signal chains, latency in digital processing can cause lip-sync issues.

31. Line level

A standard signal level for audio equipment interconnection. Consumer line level is -10 dBV (used with RCA connectors); professional line level is +4 dBu (used with XLR/TRS). Microphone-level signals must be amplified to line level before most processing equipment.

32. Lux

A unit of illuminance measuring the amount of light falling on a surface. 1 lux = 1 lumen per square meter. Classroom ambient light typically ranges from 300–500 lux; bright sunlight near windows may exceed 10,000 lux, washing out projected images.

33. MAC address

A unique hardware identifier assigned to every network interface at manufacture, expressed as six pairs of hexadecimal digits (e.g., 00:1A:2B:3C:4D:5E). The first

three pairs identify the manufacturer (OUI) and can be used to identify AV device brands on a network.

34. Omnidirectional

A microphone polar pattern that captures sound equally from all directions. Useful for conference room applications where multiple speakers are seated around a table; more susceptible to feedback than cardioid patterns when used with loudspeakers in the same room.

35. OUI (Organizationally Unique Identifier)

The first 24 bits (3 bytes) of a MAC address that identify the device manufacturer. Used during network discovery to identify AV equipment brands without knowing the device model or IP address.

36. Phantom power (+48V)

A DC voltage (typically +48 volts) delivered through an XLR cable to power condenser microphones. Provided by mixers, audio interfaces, and preamps. Dynamic microphones are not damaged by phantom power but do not use it.

37. Polar pattern

A diagram showing the directional sensitivity of a microphone — how well it picks up sound from different angles. Common patterns: cardioid (front-focused), omnidirectional (all directions), figure-8 (front and back).

38. QoS (Quality of Service)

Network configuration settings that prioritize certain types of traffic (e.g., video conferencing audio/video) over others (e.g., file downloads) to ensure consistent performance. Critical for AV-over-IP systems sharing network infrastructure with general school traffic.

39. Root cause

The fundamental reason a problem exists, as distinguished from its symptoms. In AV troubleshooting, fixing a symptom (e.g., selecting a different input) without addressing the root cause (e.g., a broken switcher preset) will allow the problem to recur.

40. Signal flow

The path an audio or video signal travels through a system, from its source through processing and distribution to the output device. Understanding signal flow is the foundation of AV troubleshooting and system design.

41. Signal integrity

The degree to which a signal accurately represents the original at the receiving end. Poor signal integrity results in image degradation, audio noise, or signal dropout, caused by excessive cable length, interference, or damaged connectors.

42. SPL (Sound Pressure Level)

A measurement of the pressure variation caused by sound waves, expressed in decibels (dB SPL) relative to the threshold of human hearing. Normal conversation is approximately 60–65 dB SPL; the OSHA-recommended limit for prolonged exposure is 85 dB SPL.

43. Subnet

A logical subdivision of an IP network, defined by a subnet mask. Devices within the same subnet communicate directly; communication between subnets requires a router. AV devices are often placed on a dedicated subnet for performance and security.

44. Telecoil (T-coil)

A small wireless receiver built into hearing aids and cochlear implants that picks up magnetic signals from a hearing loop (induction loop). Users activate T-coil mode on their device to receive the direct audio signal without background noise.

45. Throw ratio

The relationship between a projector's distance from the screen and the width of the projected image: $\text{Throw Ratio} = \text{Distance} \div \text{Screen Width}$. A throw ratio of 2:1 means the projector must be 2 feet away for every 1 foot of screen width. Short-throw projectors have ratios below 0.6:1.

46. VLAN (Virtual Local Area Network)

A logical network segmentation technique that groups devices on a shared physical switch into isolated virtual networks. AV devices placed on a dedicated VLAN are protected from general network traffic and can receive QoS prioritization.

47. WCAG (Web Content Accessibility Guidelines)

A set of internationally recognized technical standards published by the W3C that define how digital content should be made accessible to people with disabilities. WCAG 2.1 Level AA is the most commonly required compliance level for public education institutions in the United States.

48. Web interface

A browser-based configuration and monitoring page built into many networked AV devices. By entering the device's IP address in a browser, administrators can update firmware, configure settings, and monitor device status without physical access to the device.

49. XLR connector

A rugged 3-pin circular connector used for professional balanced audio connections. Pin 1 = Ground, Pin 2 = Signal (+), Pin 3 = Signal (-). Female XLR connectors are typically on inputs (microphones); male XLR connectors are typically on outputs (cables connecting to mixers).

50. Zero-latency monitoring

An audio interface feature that allows a performer or presenter to hear themselves directly through headphones without the processing delay introduced by the connected computer. Important during recording sessions and live streaming to prevent performers from hearing a distracting echo of their own voice.

APPENDIX E

Alternative Lab Options

For schools with limited equipment, internet-only access, or remote/hybrid course delivery. Each alternative is designed to achieve equivalent learning outcomes. Instructor must pre-approve selected alternative format.

How to Use This Appendix

Each lab below lists: (1) a **Virtual Simulation** option using free browser-based tools, (2) a **Video Observation** option using freely available online video content, and (3) a **Partner Institution** option for schools that can arrange a site visit to a neighboring school, university, or AV dealer demonstration room. All alternative labs produce the same worksheet deliverables as the standard lab —

only the equipment access method differs.

Lab 1 Alternative: Exploring the AV Ecosystem

Alternative Type	Resource / Method	Deliverable Same As Standard Lab?
Virtual Simulation	Use Google Street View or a virtual campus tour of a university with publicly available building walk-throughs. Document AV components visible in the environment. Combine with manufacturer product-page research to complete the inventory table.	Yes — complete standard AV Room Inventory Table and Signal Flow Template based on what is observable.
Video Observation	Watch a video tour of a technology-equipped classroom (search: "smart classroom tour" on YouTube). Pause and document all visible AV components. Many university media services channels post detailed room tours.	Yes — same inventory table; note "Video Observation" as source.
Partner Institution	Arrange a guided visit to a neighboring district's AV-equipped classroom, university media room, or AV dealer demonstration facility. Conduct the full lab in person at the partner site.	Yes — full standard lab with all worksheets.

Lab 2 Alternative: Audio Signal Chain Exploration

Alternative Type	Resource / Method	Deliverable
Virtual Simulation	Use Chrome Music Lab (musiclab.chromeexperiments.com) for frequency and waveform visualization. Use NIOSH SLM app on smartphone to measure ambient room SPL levels. Audacity can be used with a built-in laptop microphone for signal chain simulation.	Complete SPL log using smartphone measurements; complete EQ observation using Audacity with built-in mic; draw signal chain based on laptop audio architecture.
Video Observation	Watch a professional audio tutorial demonstrating gain staging, EQ, and feedback management (search: "audio signal chain explained" or "gain staging tutorial" on YouTube). Document the signal chain shown in the video.	Complete signal chain documentation table based on the video; answer reflection questions based on observed demonstrations.
Partner Institution	Visit a school music room, theatre, or any space with a mixing board setup. Conduct measurement steps only (Steps 2, 5, 6) with permission from the music educator on duty.	Partial lab — complete measurement tables; supplement with video observation for steps requiring live feedback test.

Lab 3 Alternative: Video Display Evaluation

Alternative Type	Resource / Method	Deliverable
Virtual Simulation	Use manufacturer specification sheets (freely available from projector and display	Complete comparison matrix with spec-based entries; complete throw ratio calculation using

Alternative Type	Resource / Method	Deliverable
	manufacturers) to complete the comparison matrix using published specification values rather than measured values. Clearly mark all entries as "Spec Sheet — Not Measured."	published throw ratio and a hypothetical room dimension.
Video Observation	Watch a professional display technology review or comparison (search: "projector vs flat panel classroom comparison" or "display technology review education"). Document technology characteristics observed in the video.	Complete comparison matrix based on reviewer's findings; answer all reflection questions.
Partner Institution	Visit a neighboring school, AV dealer showroom, or university AV lab that has multiple display types available for demonstration.	Full standard lab.

Lab 4 Alternative: Signal Flow Tracing and Documentation

Alternative Type	Resource / Method	Deliverable
Virtual Simulation	Use draw.io (app.diagrams.net) with the "Network" or "Electrical" shape library to create a signal flow diagram for a hypothetical classroom AV system. Use the equipment inventory from Lab 1 (standard or alternative) as your source list.	Complete signal flow diagram in draw.io (export as PDF); complete documentation table based on hypothetical system.
Video Observation	Find a video showing an	Complete signal flow

Alternative Type	Resource / Method	Deliverable
	AV system installation or a "behind the rack" tour (search: "AV rack tour" or "classroom AV installation walkthrough"). Trace the signal flow shown in the video.	documentation table and diagram based on the observed system.
Partner Institution	Visit a school AV closet or equipment rack with permission. Document cable runs visible in the rack and to wall plates.	Full standard lab limited to accessible cable segments; draw.io diagram for complete system.

Lab 5 Alternative: Network Discovery for AV

Alternative Type	Resource / Method	Deliverable
Virtual Simulation	Use the Cisco Packet Tracer network simulation tool (free with registration at netacad.com) to build a simulated AV network with assigned IP addresses. Alternatively, use a home router's connected devices list (instructor's home or lab router — not school production network) to practice IP documentation.	Complete network inventory table from simulation or router interface; draw network topology diagram from simulation view.
Video Observation	Watch a video demonstrating IP scanner use in a professional AV or IT context (search: "Angry IP Scanner tutorial" or "AV network discovery"). Document the process and answer reflection questions based on the demonstration.	Complete reflection questions; write a narrative description of the network discovery process in lieu of the measured inventory table.
Partner Institution	Work with the school IT	Complete network

Alternative Type	Resource / Method	Deliverable
	department to view (not modify) the AV device list in the school's network management dashboard during a supervised session. Document observable information only.	inventory table from school network documentation with IT supervisor review.

Lab 6 Alternative: Control System Interface Evaluation

Alternative Type	Resource / Method	Deliverable
Virtual Simulation	Evaluate a virtual control system interface: many manufacturers (Crestron, Extron, Q-SYS) provide online demo interfaces or simulator apps. Alternatively, evaluate any complex software interface (a video conferencing platform's meeting controls page) as a comparable heuristic evaluation target.	Full standard heuristic evaluation form and accessibility checklist for the simulated or substitute interface.
Video Observation	Find a video demonstrating a control system touchpanel in use (search: "Crestron touchpanel demo classroom" or "Extron control system walkthrough"). Evaluate the interface as shown in the video using the heuristic evaluation form.	Complete heuristic evaluation form based on video observation; note "Video Observation" as evaluation method.
Partner Institution	Visit a school, university, or AV dealer with a control system touchpanel available for hands-on evaluation.	Full standard lab.

Lab 7 Alternative: Video Conferencing Room Audit

Alternative Type	Resource / Method	Deliverable
Virtual Simulation	Conduct a self-evaluation video conferencing session using your own device (laptop with built-in camera and microphone). Record a 3-minute session, then review the recording critically as both in-room and remote participant. Evaluate the recording using the audit form.	Full audit form completed based on self-recorded session review; note "Self-evaluation via recording" as method.
Video Observation	Find a publicly available demonstration of a professional video conferencing room (search: "hybrid classroom setup demo" or "Teams Rooms demonstration"). Evaluate the system shown using the audit form.	Complete audit form based on the demonstration video; answer all reflection questions.
Partner Institution	Visit a neighboring school, university library room, or business conference room equipped with a video conferencing system and arrange a brief remote test session.	Full standard lab.

Lab 8 Alternative: Classroom AV Design Sketch

Alternative Type	Resource / Method	Deliverable
Virtual Simulation	Use Floorplanner.com	Exported floor plan PDF

Alternative Type	Resource / Method	Deliverable
	(free browser-based tool) or RoomSketcher (free tier) to draw the 30×40 ft room to scale. Place generic AV components from the furniture/objects library. Export as PDF or screenshot. The provided scenario dimensions and requirements apply exactly as written.	plus completed bill of materials and sight line calculation template.
Paper-Only Option	Use standard graph paper (4 squares per inch is ideal). Scale: 1 square = 2 ft. This is the same as the standard lab — no digital tools required. The grid in this workbook may be used directly.	Completed workbook grid with annotated legend, plus bill of materials and sight line calculations.
Partner Institution	Measure and sketch a real existing classroom at a partner school for comparison, then design the hypothetical 30×40 room as an improvement exercise based on what you observed.	Both the observed room sketch and the designed room sketch; comparative analysis in reflection responses.

Lab 9 Alternative: Digital Signage Content Planning

Alternative Type	Resource / Method	Deliverable
Virtual Simulation	This lab is largely software-based and requires no physical AV equipment. Use Canva (free at canva.com), Google Slides, or PowerPoint on any device. The full standard lab procedure applies — no substitution needed for most schools.	Full standard lab deliverables: content plan, calendar, two sample slides, accessibility checklist, and policy outline.

Alternative Type	Resource / Method	Deliverable
Video Observation	If presentation tools are unavailable, observe and evaluate an existing digital signage system in a public location (school, library, transit station, or retail environment) and document what you observe against the content plan requirements and accessibility checklist.	Completed content plan and accessibility checklist based on an existing observed system; explain what you would change and why in the reflection questions.
Partner Institution	N/A — Lab 9 is self-contained and software-only. No partner institution visit required.	Full standard lab as described above.

Lab 10 Alternative: AV Troubleshooting Challenge

Alternative Type	Resource / Method	Deliverable
Virtual Simulation	Use a video conferencing platform (Zoom, Teams) as a simulated AV system. Instructor introduces faults by disabling camera, muting microphone remotely, presenting incorrect share, or changing bandwidth settings. Student troubleshoots within the platform environment following the same systematic methodology.	Full troubleshooting log using platform faults; fault report and prevention procedure completed for software-layer faults.
Case Study Option	Instructor provides 3 written AV fault scenario descriptions (including symptoms, system diagram, and contextual information). Student applies the	Full troubleshooting log, fault report, and prevention procedure completed analytically; note "Case Study — Analytical Method" on submission.

Alternative Type	Resource / Method	Deliverable
	troubleshooting methodology analytically, documenting the process as if conducting live troubleshooting.	
Partner Institution	Visit a neighboring school's AV-equipped classroom and have the host AV coordinator or IT staff introduce controlled faults (with their permission and cooperation) for hands-on troubleshooting practice.	Full standard lab.

Lab 11 Alternative: Accessibility Audit

Alternative Type	Resource / Method	Deliverable
Virtual Campus Tour	Many universities and schools offer public virtual campus tours. Use a virtual tour to conduct a visual accessibility audit of the spaces shown, completing as many of the 50 checklist items as are observable. Note "Virtual Observation — Limited Scope" for items that cannot be assessed remotely.	Partially completed 50-item checklist with clearly marked observable and non-observable items; full remediation matrix for observable items; administrative memo template.
Video Observation	Find a video documenting an accessibility audit or assistive technology demonstration in an educational setting. Evaluate the space shown against the 50-item checklist.	Completed checklist items where observable; reflection questions answered fully.
Home/Community Audit	With instructor approval, conduct the audit on a public community space (library meeting room,	Full standard lab deliverables for the community space audited.

Alternative Type	Resource / Method	Deliverable
	community center classroom) rather than a school room. Apply the same 50-item checklist and deliver the same administrative memo to a hypothetical administrator.	

Lab 12 Alternative: Emerging Technology Exploration

Alternative Type	Resource / Method	Deliverable
Remote Presentation	Lab 12 requires only a computer and internet access — no AV equipment. The full standard lab applies for all students regardless of equipment availability. If a live class presentation is not possible, submit a recorded video presentation (5 minutes, same slide requirements) in lieu of a live presentation.	Research template, recorded or live presentation, peer evaluation forms (completed asynchronously via shared document), written summary.
Written Report Option	If presentation is not feasible in any format, submit an expanded written report (3–4 pages) covering all fields of the research template with additional depth in the educational use case and equity analysis sections.	Written report covering all research template fields; peer review replaced by instructor evaluation using the same criteria.

 **Instructor Note: Grading Alternative Labs**

Alternative lab formats are graded using the same rubric as standard labs. Where a deliverable cannot be produced due to the nature of the alternative (e.g., SPL measurements from a virtual simulation), instructors should accept a clearly labeled analytical estimate with written justification in lieu of a measured value. The quality of the student's reasoning and documentation process is the primary assessment target, not access to specific equipment.

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